KARTHIKA SENTHIL KUMAR

AI&ML ENGINEERING

7200671716 karthikas2566@gmail.com Erode-11, India GitHub LinkedIn Leetcode Portfolio

ABOUT ME

Computer Science (AI&ML) student with a focus on Python, Machine Learning, and IoT-based real-time systems, with additional experience in full-stack web development. Through hands-on projects, a published patent, and industrial data-analytics work, I enjoy turning innovation and technology into practical, real-world solutions. Keen to learn from industry best practices and contribute to the software-defined, AI-enabled future of mobility through the **Visteon SCHOLAR Program**.

EDUCATION

KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY, COIMBATORE **2024-2028**
BE.CSE(AI&ML) - 8.45

VELLALAR MATRIC.HR.SEC.SCHOOL, ERODE **2021-2024**
HSC (Maths & Computer Science) - 88%

PROJECTS

Smart Home Alert System (Patent Published) **2025**
- Developed a smart wearable wristband that delivers real-time household safety alerts
- Integrated an OLED display and vibration motor for instant on-device alerts and feedback
- Connected with IoT sensors for real-time alerts on door bell, gas leakage, water overflow, and temperature variation

Employee Performance Prediction - ML **2026**
- Built an ML model using data such as attendance, hours worked, productivity, and history of past performance.
- Covered the end-to-end pipeline: data preprocessing, feature selection, and model training.

INTERNSHIPS

Data Analyst / Mechanical Intern - LITC, Chennai (June 2026 - July 2026)
- **Lennox International, Inc.** is a global HVAC company specializing in heating, ventilation, air conditioning, and climate-control solutions.
- Developed a Matrix Profile-based web application for **time-series pattern and anomaly detection** using STAMP and MASS.

Data Management - Tekfilo, Coimbatore (Dec 2025 - Jan 2026)
- **Software Development Intern:** Developed a Gold Transaction Data Management System using **Spring Boot, Hibernate, and Angular**.
- Developed a Dynamic Excel Upload collaboratively and practically with team members.

Frontend Development - CodeAlpha (Dec 2025 - Jan 2026)
- Built portfolio, calculator, and code editor image gallery. Learned and practiced HTML, CSS, and JavaScript

Infosys Springboard - Virtual Intern, Batch 5 (Oct 2026)
- Selected for the **Infosys Springboard Internship 7.0** virtual internship program.

ACHIEVEMENTS

- **Published a patent** for "Smart Home Alert System" on 6 June 2025 (India) - **Author.**
- **Best Innovators** with cash prize at the international level 360 degree Hackathon 3.0 in **EC Learnix**.
- **Prize with cash award** for (Smart Home Alert System) in Karpagam Institute of Technology, Coimbatore.
- **Price with Case Award** for (Smart Home Alert System) at CMR Institute of Technology, Bangalore.
- **Research Paper:** Conference paper on **Context-Aware Dynamic Skill Gap Analysis using Machine Learning** — under preparation for conference publication.
- Solved **500+ problems on SkillRack**, building strong fundamentals in Python, Java, and MySQL.
- **Token of Appreciation** from Zoho Apptics & Codeflakes Studio Co-Founder for the **Smart Home Alert System** and developing an **iOS Workout Trainer App** during their iOS development.
- Visited the **IIT Madras Heritage Centre** and the IIT Palakkad HPC Centre.

CERTIFICATIONS

- **Machine Learning Using Python** - Simplilearn
- **NPTEL Certificate (Elite)** - Industry 4.0
- **NPTEL Certificate (Elite)** - LLM'S
- **Bajaj Manufacturing Systems** - Certification
- **Programming Certifications** - C, C++, Python, Java - Spoken Tutorial Project, IIT Bombay
- **AWS Fundamentals for Cloud Practitioners** - Nminds Academy
- **Coding & Programming** - Samsung Innovation Campus
- **CodeChef Certifications** - Learn Java / Practice Java (500-1000 difficulty-rated problems, strings and math)
- Completed Certification Courses on **Infosys Springboard**

SKILLS

- **Languages:** Python, Java, C++
- **Machine Learning:** TensorFlow, Scikit-learn, NumPy, Pandas, KNN, Linear & Logistic Regression
- **AI/ML Concepts:** Anomaly Detection, Pattern Recognition, NLP, Recommendation Systems
- **Data Analytics:** Data Preprocessing, EDA, Feature Engineering, Time-Series Analysis
- **IoT & Embedded Systems:** Sensor Integration, Real-Time Alerts, Wearable Prototyping
- **Backend:** Spring Boot, Hibernate, Node.js, FastAPI
- **Web Development:** HTML, CSS, JS, React
- **Databases:** MySQL, PostgreSQL, MongoDB, Firebase
- **Tools & Technologies:** Git, GitHub, Maven, NPM, Excel